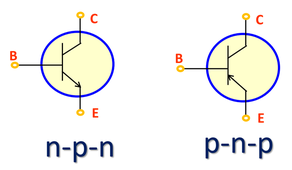
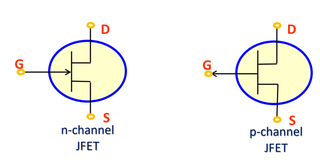
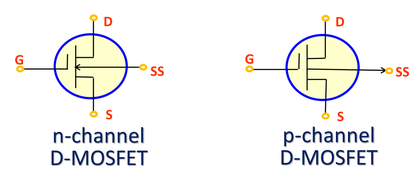
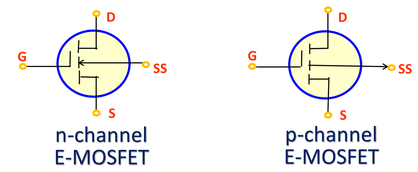
# Comparision of Transistors (Summary of BJT & FET)

| Parameters | BJT (Bipolar Junction Transistor) | FET (Field Effect Transistor) | | |
|---|---|---|---|---|
| | | JFET | MOSFET | |
| | | | Depletion Type MOSFET | Enhancement Type MOSFET |
| Types & Symbol |   |   |   |   |
| Operation | $I_E = I_B + I_C$ <br> 2 % 98 % | (i) $V_{GS} = 0 \Rightarrow I_D = I_{DSS}$ <br> (ii) With small $-V_{GS}$ <br> $\Rightarrow I_D$ reduced <br> (iii) With large $-V_{GS}$ <br> $\Rightarrow I_D = 0$ | (i) $V_{GS} = 0 \Rightarrow I_D = I_{DSS}$ <br> (ii) With Increase $-V_{GS}$ <br> $\Rightarrow I_D$ reduced <br> (iii) With $+V_{GS}$ <br> $\Rightarrow I_D$ rapid increase | (i) $V_{GS} = 0 \Rightarrow I_D = 0$ <br> (ii) With increase $+V_{GS}$ <br> $\Rightarrow$ induced channel, $I_D$ increased <br> (iii) With Const $V_{GS}$ inecreced $V_{DS}$ <br> $\Rightarrow I_D$ = Const and saturation region |
| Isolation of controlling terminal from thr device structure | Base is not isolated | Gate is not isolated | Sio2 layer between gate and Channel | Sio2 layer between gate and substarte |
| Input Resistance | Low | High | Very High | Very High |
| Noise produced | High | low | Very low | Very low |
| Size | Bigger than JFET | Small | Very Small | Very Small |
| Switching Speed | Low | High | Very High | Very High |
| Output Current | $I_C = \beta_{dc} I_B$ | $I_D = I_{DSS}\left[1 - \frac{V_{GS}}{V_P}\right]^2$ | | $I_D = K[V_{GS} - V_T]^2$ |
| Current Gain | For CB --> $\alpha_{dc} = \frac{I_C}{I_E}$ <br> For CE --> $\beta_{dc} = \frac{I_C}{I_B}$ <br> For CC --> $\gamma_{dc} = \frac{I_E}{I_B}$ | $\mu = \frac{\Delta V_{DS}}{\Delta V_{GS}}$ , $g_m = \frac{\Delta I_D}{\Delta V_{GS}}$ | | |
| Terminal | Base, Emitter, Collector | Gate, Source, Drain | | |
| Confuguration | CB, CE, CC | CG, CS, CD | | |
| Device Type | Bipolar | Unipolar | | |
| Control Element | Current Control Device | Voltage Control Device | | |
| Temperature Co-efficent of Resistance | Negative | Positive | | |
| Thermal Runaway | Damage the BJT | Does not exists in FET | | |
| Thermal Stability | Less | High | | |
| Characteristics | Input & Output Characteristics | Drain Characteristics (Drain Curve) <br> Transfer Characteristics (Transconductance Curve) | | |
| Relation between input and output | Linear | Non-Linear | | |

Prepared by :- Asst. Prof. Jayesh Prajapati